

Subdomain Discovery_ Passive+Active Sub Discovery

Name: FIREWALL FAIRY

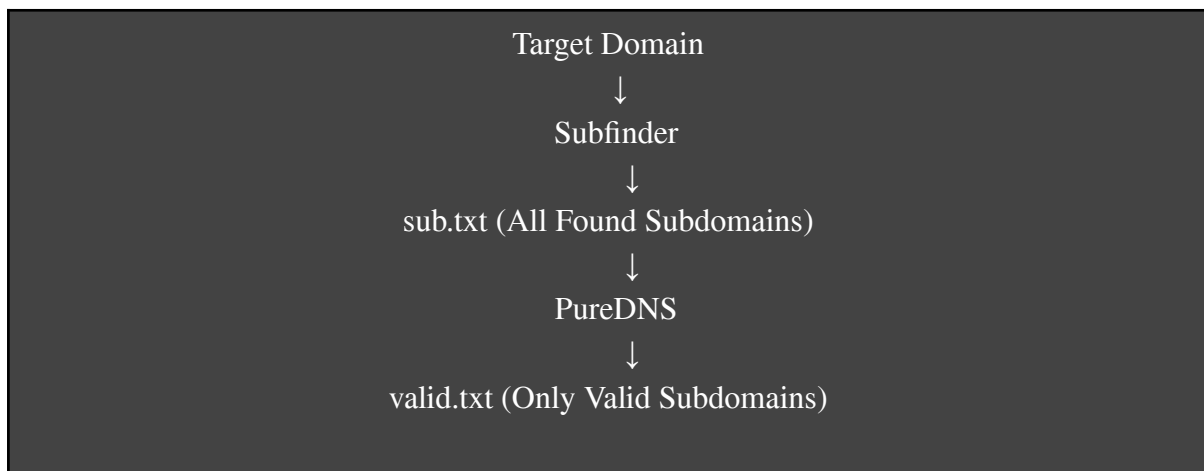
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1. Overview

Subdomain Discovery is the process of finding subdomains of a target domain. During this practical, I used two tools:

- **Subfinder** – Finds subdomains from passive online sources.
- **PureDNS** – Checks which discovered subdomains are actually valid and resolve through DNS.

Workflow



2. Objective

1. Find subdomains of a target domain.
2. Save all discovered subdomains.
3. Verify which subdomains are valid.
4. Save only valid subdomains for the next reconnaissance step.

3. Installation

Install Subfinder

```
sudo apt install subfinder
```

```
(kali㉿kali)-[~]  
$ sudo apt install subfinder  
Installing:  
subfinder
```

Verify installation

```
subfinder-h
```

```
(kali㉿kali)-[~]  
$ subfinder -h  
Subfinder is a subdomain discovery tool that discovers subdomains for websites by using passive online sources.
```

```
(kali㉿kali)-[~]  
$ subfinder -ls  
go install github.com/d3mondev/puredns/v2@latest
```

```
(kali㉿kali)-[~]  
$ subfinder -u https://projectdiscovery.io  
remote: Enumerating objects: 1231, done.  
remote: Counting objects: 100% (715/715), done.  
remote: Compressing objects: 100% (715/715), done.  
remote: Total 1231 (delta 100), reused 0 (delta 0), pack-reused 1231  
projectdiscovery.io
```

4. Install PureDNS

First, install Go.

```
sudo apt update  
sudo apt install golang-go
```

```
(kali㉿kali)-[~]  
$ sudo apt update  
sudo apt install golang-go  
[sudo] password for kali:
```

Install PureDNS.

```
go install github.com/d3mondev/puredns/v2@latest
```

```
(kali㉿kali)-[~]  
$ go install github.com/d3mondev/puredns/v2@latest
```

Install MassDNS (required by PureDNS).

```
git clone https://github.com/blechschmidt/massdns.git  
cd massdns  
make  
sudo make install
```

```
(kali㉿kali)-[~]  
$ git clone https://github.com/blechschmidt/massdns.git  
cd massdns  
make  
sudo make install
```

5. Issue I Faced

After installing PureDNS, I tried:

```
puredns--version
```

Error:

```
puredns: command not found
```

```
(kali㉿kali)-[~]  
$ puredns resolve sub.txt -r ~/.config/puredns/resolvers.txt  
puredns: command not found
```

Reason

The Go binary folder (`~/go/bin`) was not added to the system PATH.

Solution

Check if PureDNS was installed.

```
ls ~/go/bin
```

Run it directly.

```
~/go/bin/puredns--version
```

Output

```
puredns version v2.1.1
```

Add Go binary to PATH.

```
echo 'export PATH="$HOME/go/bin:$PATH"' >> ~/.zshrc  
source ~/.zshrc
```

Verify.

```
which puredns
```

Result

```
/home/kali/go/bin/puredns
```

Now the command works.

```
puredns--version
```

Output

```
puredns version v2.1.1
```

```
(kali@kali)-[~]  
$ puredns --version  
puredns version v2.1.1
```


Step 2 – Save the Results

Command

```
subfinder-d hackerone.com > sub.txt
```

Purpose

Save all discovered subdomains into a file.

Check the file

```
cat sub.txt
```

Result

```
sub.txt contains 24 discovered subdomains.
```

Step 3 – Download Resolver List

PureDNS requires DNS resolvers.

Create the configuration folder.

```
mkdir -p ~/.config/puredns
```

Download the resolver list.

```
wget <https://raw.githubusercontent.com/trickest/resolvers/main/resolvers.txt> \  
-O ~/.config/puredns/resolvers.txt
```

Check the file.

```
head ~/.config/puredns/resolvers.txt
```

```
(kali㉿kali)-[~]  
$ head ~/.config/puredns/resolvers.txt  
99.99.99.193  
99.92.207.51  
99.83.147.72  
99.79.50.252  
99.79.182.170  
99.42.199.193  
98.26.23.128  
98.213.193.14  
98.213.192.240  
98.213.192.223
```

Step 4 – Validate Subdomains

Command

```
puredns resolve sub.txt \  
-r ~/.config/puredns/resolvers.txt
```

PureDNS checked all discovered subdomains and removed invalid ones.

Result

- 24 subdomains received from Subfinder.
- 10 valid subdomains found after DNS validation.

Example

```
www.hackerone.com  
api.hackerone.com
```



```
docs.hackerone.com
support.hackerone.com
```

```
Found 10 valid domains:
b.ns.hackerone.com
www.hackerone.com
docs.hackerone.com
api.hackerone.com
a.ns.hackerone.com
mta-sts.hackerone.com
mta-sts.managed.hackerone.com
gslink.hackerone.com
support.hackerone.com
mta-sts.forwarding.hackerone.com
```

Step 5 – Save Valid Subdomains

Command

```
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt \
-w valid.txt
```

Check the output.

```
cat valid.txt
```

Result:

The file contains only valid subdomains.

```

(kali㉿kali)-[~]
$ cat valid.txt
a.ns.hackerone.com
b.ns.hackerone.com
www.hackerone.com
api.hackerone.com
mta-sts.hackerone.com
mta-sts.managed.hackerone.com
docs.hackerone.com
support.hackerone.com
gslink.hackerone.com
mta-sts.forwarding.hackerone.com

```

7. What I Learned

- Subfinder quickly finds subdomains using passive online sources.
- Not every discovered subdomain is valid.
- PureDNS checks which subdomains actually resolve through DNS.
- Saving valid subdomains makes the next reconnaissance steps more accurate.
- Fixing the PATH variable was necessary to run PureDNS from anywhere in the terminal.

9. Important Commands (Cheat Sheet)

```

# Find subdomains
subfinder -d hackerone.com

# Save results
subfinder -d hackerone.com > sub.txt

# Check version
puredns --version

# Validate subdomains
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt

# Save valid subdomains
puredns resolve sub.txt \
-r ~/.config/puredns/resolvers.txt \

```

```
-w valid.txt
```

```
# View results
```

```
cat sub.txt
```

```
cat valid.txt
```